Daniel V. Rosamond

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EDUCATION

University at Buffalo, Buffalo, NY
Bachelor of Science, Mechanical Engineering, Minor in Robotics

TECHNICAL SKILLS

- CAD & Analysis: Fusion 360, SolidWorks, FEA, Modal Analysis, Dynamic Event Simulation
- Robotics & Embedded: ESP32, Arduino, Raspberry Pi, PCB design, Motor/ESC/Sensor integration
- Programming: C/C++, Python

ENGINEERING EXPERIENCE

American Society of Mechanical Engineers (ASME), University at Buffalo

Engineering Intern, *EMPEQ, Ithaca, NY* — March 2026 — Present

- Supporting engineering on a **Direct-to-Phase II SBIR contract (AFWERX / U.S. Air Force)** focused on hardware supply chain modernization for defense and utility sectors; **NYSERDA** internship
- Cataloguing nameplate and technical spec data from industrial, defense, and HVAC equipment via structured EL workflow to build the supply chain intelligence database

Rover Systems Lead – ASME IAM3D 2026 — August 2025 — Present

- Led a **50-member, multi-subteam engineering program** (drivetrain, payload, chassis, electronics, testing) to design a fully 3D printed excavation rover within a 30x30x30cm constraint
- Engineered parametrized, **sand-optimized treads, drivetrain,** validated through sandbox, FEA
- Architected the rover's complete electronics system, **designing a custom PCB and integrating an ESP32** across dozens of IO channels for motor control, sensing, power distribution
- Performed **extensive FEA and modal analysis** across rover components; tuned camera gimbal's natural frequency to servo vibration modes using flexible materials, eliminating image distortion
- Authored **160-page engineering report** covering simulation, electronics, DFAM/DFMA, and testing

Fabrication Lab Manager – ASME UB — January 2025 — Present

- **Managed** student fabrication lab (FDM, CNC, laser, GF/CF/KF Layups) supporting **130+ active members**
- **Trained and certified** 3D-printer operators to perform standardized preventive maintenance, increasing machine uptime from **~20% to >60%**
- Built **Raspberry Pi lab display system**; integrating print monitoring capabilities
- Secured Autodesk ($700) and Red Bull (100 units/month), expanded lab capacity by ~300%

TECHNICAL PROJECTS

FPV Drone – ASME IAM3D 2025 — September 2024 — April 2025

- Co-led development of a **fully 3D-printed FPV payload drone** within a 30x30x30cm limit using DFAM, print-in-place joints, friction-fit mechanisms; achieved **>2:1** thrust-to-weight, ~16 min flight time, and **~4.9 lb** measured payload capacity to complete the required **100g pickup-transport-drop course**
- Developed PLA/TPU struct-core elastomer frame that survived **13 impacts at 40–60 mph** in crash testing

50lb Combat Robot – ASME EBoard Battlebot — September 2025 — February 2026

- Led design and machining of a 50lb combat robot in team of three; placed in **top four** at UB EWeek 2026
- Designed a weapon system operating at **25kJ** (≈small car crash energy) within safety and competition constraints. Simulated weapon inertia, stresses, and drivetrain loads

ADDITIONAL INFORMATION

Fire Safety Technician, *UB EHS* — August 2025 — April 2026

- Performed campuswide NFPA/OSHA inspections and preventive maintenance on fire-safety systems; maintained compliance records and asset logs.

Languages: English, Polish, Latin (advanced)
Volunteer college essay tutor, admitting students into top universities including Vanderbilt, Emory